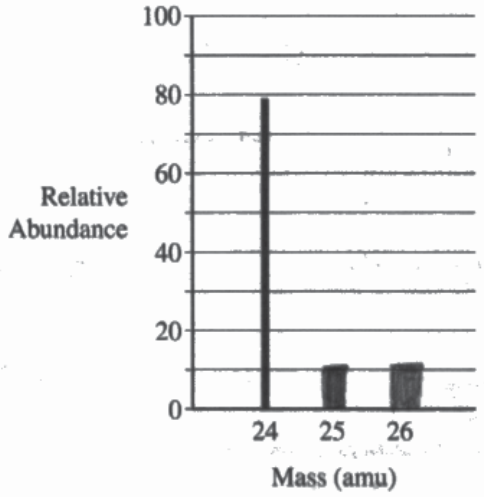
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Part A

Question 1

i.



ii. Both magnesium-25 and magnesium-26 have 12 protons. Magnesium-26, however, has 14 neutrons while magnesium-25 has 13 neutrons. This additional neutron is responsible for the difference in mass.

Question 1 is continued on the next page.

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Question 1

Part B

From Coulomb's Law, $F_{coulombic} \propto \frac{q_1 q_2}{r^2}$.

i. Mg^{+2} has a larger oxidation number of +2 compared to +1 for Na^{+1}. From the numerator of Coulomb's Law, a larger charge means a larger force. Mg^{+2} has a larger ~~positive charge~~ than Na^{+1}, meaning Na^{+1} will have a weaker attraction to the partial negative charge of the O in water than Mg^{+2}.

ii. From the denominator of Coulomb's Law, a larger separation means a smaller Coulombic force or attraction. Both Na^{+1} and Mg^{+2} have 10 electrons, but Na^{+1} has one less proton than Mg^{+2} (11 vs. 12), so it will have less proton pull on each individual electron, resulting in a larger electron cloud and larger ionic radius for Na^{+1}. This larger radius for Na^{+1} results in a larger separation between the Na^{+1} ion and water molecule, resulting in a weaker attraction than with Mg^{+2}.

Part C

$pOH = -\log[OH^-] = -\log(2.8 \times 10^{-4}) = 3.55$

$pH = 14 - pOH = 10.45$

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Question 1

Part D

$$\frac{\frac{1.85\times10^{-3}\text{ mol } Mg(NO_3)_2}{1\text{ L}}\times 0.0350\text{ L}}{0.085\text{ L}} = 7.62\times10^{-4}\text{ M } Mg(NO_3)_2$$

$[Mg(NO_3)_2] = [Mg^{+2}] = 7.62\times10^{-4}\text{ M } Mg^{+2}$

Part E

i. $K_{sp} = [Mg^{+2}][OH^-]^2$

ii. $Q = (7.62\times10^{-4})(1.65\times10^{-4})^2 = 2.07\times10^{-11}$

iii. Since $Q > K$, the reaction will shift left to approach equilibrium, producing $Mg(OH)_{2(s)}$. So yes, a precipitate should form.

Question 1 is continued on the next page.

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Question 1

Part F

The amount of undissolved $Mg(OH)_{2(s)}$ decreases. The added $HNO_{3(aq)}$ completely dissociates into $H^+_{(aq)}$ and $NO^-_{3(aq)}$. The increased $[H^+]$ decreases the $[OH^-]$ ($K_w = [H_3O^+][OH^-]$). The decreased $[OH^-]$ causes the equation from Part E to shift right from Le Chatelier's Principle, decreasing the amount of $Mg(OH)_{2(s)}$ left undissolved.

Continue to Question 2.